COMSATS University Islamabad, *Lahore Campus*

Department of Computer Engineering

Quiz 1 – SPRING 2026

Course Title:	Digital System Design		Course Code:	CPE244	Credit Hours: 4(3,1)
Course Instructor:	Dr. Engr. M Babar Ali		Program Name:	BSc in Computer Engineering	
Semester:	7th	Batch:	FA23	Section:	A/B
Date:		**Time Allowed:**	**15 minutes**	**Maximum Marks:**	**10**
Student's Name:		Reg. No.	CUI/ - - /LHR		

Important Instructions / Guidelines:

- Write your name and registration number clearly in respective fields.
- Mobile phones are not allowed in the classroom.
- Do not give multiple answers to a question.

Question: [CLO1, PLO3 – C5] **(Marks 5+5)**

1. A battery-level sensor outputs the remaining charge as a 4-bit binary number, where 0000 represents an empty battery and 1111 represents a fully charged battery. ***Design*** a circuit that turns on a "Low Battery" indicator as a single bit output when the charge drops below level 6 and a "Critical Battery" indicator as another single bit output when the charge drops below level 3.
2. Draw Timing diagram for the following FSM for the inputs given.



